

Immersive Technologies

Deadline	Friday 24 th April 2026 before 4pm
Submission Method	CANVAS Submission (Online) (A link will be made available closer to the submission date)
Tutors	Badheer Ballam

Module Title	Immersive Technologies	Level / Semester	Level 5
Module Code	CMS22101	Credit / Weighting	20
Module Leader	Dr Nosheen Gul	Date of Issue	23-09-2025

Immersive Technologies



Project Brief

INTRODUCTION

The convergence of physical and virtual reality is taking place. Modern societies are spending more and more time “living” via a 2D digital screen. We communicate, manage our finances, order food, find accommodation and even seek lasting emotional relationships via these digital screens. This is now becoming the norm of society.

The next step of this digital interaction will take place in the 3D world – whether that be via a VR headset, AR glasses or even AR contact lenses. The world’s largest companies are investing heavily into mixed reality and its current and future applications. Students that are proficient in developing mixed reality technologies and applications will gain a great advantage in the upcoming next stage of “Spatial computing”.

This unit will introduce students to the fundamental concepts of Mixed reality (XR). Students will explore industry specific use case and identify suitable areas for development. Students will develop proficiency in; 3D content creation, Optimization for devices, applying physics to increase immersion and developing interactivity by utilising visual scripting for rapid prototyping. In addition to the visual dimension, this unit will explore how speech recognition can be incorporated into this new reality with an emphasis on voice user interface.

ASSESSMENT PROJECT BRIEF – Breakdown / details

Objectives:

1. Create a functional Augmented Reality App for a selected industry.
2. This App should add clear value for the end user
3. Showcase your understanding of Mixed Reality /VR best practices.

Sample Topics:

- Education: AR in the classroom, Teaching Biology with AR eg. Visualise the human anatomy.
- Retail: AR in shopping, IKEA style furniture previewer, Sneakr AR – Preview pre purchase stage.
- Design: AR in Design development, Automobile industry, BMW AR, Aerospace, Large scale industries.

SUBMISSION

You are asked to submit the following work:

Project Presentation, digital media piece (5 minutes)

MS PowerPoint file (PPT) – Explaining the rationale behind the project and design choices. Video demonstration of AR App in use with correct tracking and data.

Project Report (500 - 1000 words) containing:

- 1) Demonstrate how your Augmented Reality App may solve certain issues for the selected Industry (Health/Retail/etc).
- 2) Conduct research to identify the pain points (aka opportunities) within the selected industry then formulate

solutions.

3) Research and evaluate the key elements of a successful VR app/experience. Include background research and outline your project approach.

4) Technical documentation about your project presentation (hardware and software used)

5) Include images and video documentation (max. 5 minutes) of your project.

Assessment Criteria:

1. Ability to evaluate the key elements of a successful AR app/experience. (10%):

2. Knowledge of different immersive media platforms and techniques (10%):

3. Demonstrating the key elements of a successful Augmented reality App/Experience (30%):

4. Familiarity with AR techniques and knowledge of the principles of spatial computing. (10%):

4. Clarity, structure, and professionalism of the written report. (30%):

5. Practical Application and Future Directions (10%):

Timeline:

Week 1 – Introduction to Mixed Reality. History of CGI and VR. Project brief and deliverables.

Week 2 – Find the pain points in a selected industry. Unity Visual Scripting for rapid prototyping.

Week 3 – Unity Visual Scripting for rapid prototyping

Week 4 – Playmaker - Create an interactive AR Scene with Logic.

Week 5 – Playmaker - AR in Unity with Logic. Create an functional AR Scene

Week 6 – Playmaker - Grounding your scene with Physics, Gravity, and rigid bodies. Formative assessment.

Week 7 – Design your AR app, Source 3D Models – 3D Scan objects and optimise for handheld devices.

Week 8 – Design your AR app, hands-on session with individual feedback.

Week 9 – 11 - Create your AR piece, a refining session with individual feedback.

Week 12 - Final project presentation and project portfolio feedback session.



Essential Reading List/Learning Resources to refer to from the module specification – Harvard referenced

Essential reading/Learning resources: (Provide links/details on Canvas module page)

Shead, Mark. Understanding State Machines, 2018

Linowes, Jonathan. Unity 2020 Virtual Reality Projects, 2020

Recommended reading/Learning resources: (Provide links/details on Canvas module page)

McErlean, Kelly. Interactive Narratives and Transmedia Storytelling: Creating Immersive Stories across New Media Platforms. Focal Press, 2018.

Case Studies:

[VR in business Oculus: Johnsons and Johnson](#)

[Unity AR - Case Study Wallace & Gromit](#)

[Digital Domain's "Digital Doug"](#)

Other useful information

Physical Resources:

Laptop or Desktop to design and create AR App.

University Library - Research case studies and selected areas of Interest.

USEFUL WEBSITES:

[The first 3D software in History](#)

[Intro to Playmaker](#)

[Best Practices](#)

[How Virtual Reality Became a Reality](#)

[Inside The Metaverse: What Does The Future Of Virtual Reality Feel Like?](#)

[Unity VR - YouTube](#)

[What I Learned from Spending a Week in Virtual Reality | Jak Wilmot](#)

[How immersive technologies \(AR/VR\) will shape our future | Dinesh Punni](#)

Software:

Software: Unity

Software: Playmaker – Visual Scripting

Hardware: Android Smartphone/Tablet

Assessment and tutor feedback date	Grades and feedback will be released either within or up to a maximum of 20 working days after the assessment submission date.
Academic Misconduct	As specified in our General Academic Regulations, Academic Misconduct (e.g., plagiarism, inappropriate use of AI applications and tools, collusion, fabrication of information, contract cheating etc) will be dealt with formally following our procedure for Academic Misconduct. This means for example, close similarity between students' assessments, suspected use of AI, or third-party completion of assessed work and assessments that are not referenced appropriately are likely to lead to an investigation. The penalty for academic misconduct is a fail grade with an additional penalty applied according to the gravity of the case. If you have any doubts about how to avoid plagiarism, inappropriate use of AI or how group work assessments are submitted to recognise individuals' work and contribution, please discuss this with the module tutor or Course Leader. When you submit your assessments, you will be asked to complete/sign a declaration that it is your own work.
Late submissions	You can submit work up to 24 hours after the module deadline. This work will be capped as a pass grade unless you have a 5-day approved extension or other approved Extenuating Circumstances (ECs). Please see Academic Regulations below for further details.
Academic Regulations 25/26	To pass this module, you should aim to submission work before the deadline shown within this brief. However, if you fail to submit the work you will be offered 2 further retrieval opportunities. The grade for both retrieval opportunities will be capped unless you have approved Extenuating Circumstances. You will be notified of a new deadline date for the retrieval opportunities via a 'Retrieval Module brief'. How to apply for a 5 day extension? You can apply for a 5-day extension for any module(s) during the year. This must be done before the submission deadline for each module. Click here to find 5 day extension application How to apply for Extenuating Circumstances? You can apply for Extenuating Circumstances before the deadline or within 5 days of the deadline: Click here to find EC application Note: You can apply for Extenuating Circumstances for any module, except where the assessment is 'ungraded' and marked as Pass/Fail' only. For further information please use this link to access General Academic Regulations



**Other
requirements
for submission**

Submission of report via Turn-it In on Canvas

Module learning outcomes		Assessment Criteria	Relevant Core Competency
LO_01	Critically evaluate the design and technical success of existing AR applications	Explore Research and Analysis Ability to evaluate the key elements of a successful AR app/experience.	Critically Appraise
LO_02	Compare the technical features and limitations of various immersive media platforms	Experimentation Communication Knowledge of different immersive media platforms and techniques	Distinguish / Categorize
LO_03	Develop a functional AR prototype that demonstrates effective user experience principles	Create Technical Competence Demonstrating the key elements of a successful Augmented reality App/Experience	Develop / Construct
LO_04	Apply core spatial computing principles and techniques to AR development.	Practical application Familiarity with AR techniques and knowledge of the principles of spatial computing.	Apply / Explain



Assessment Methods and Tasks		
Assessment tasks	Submission Method	Weighting %
Project Presentation, digital media piece (5 minutes)	In class presentation	50%
Write a report (500 - 1000 words) containing: 1) Demonstrate how your Augmented Reality App may solve certain issues for the selected Industry (Health/Retail/etc). 2) Conduct research to identify the pain points (aka opportunities) within the selected industry then formulate solutions. 3) Research and evaluate the key elements of a successful VR app/experience. Include background research and outline your project approach. 4) Technical documentation about your project presentation (hardware and software used) 5) Include images and video documentation (max. 2 minutes) of your project. Assessment Rubric (attached separately)	Upload to Aula in Pdf / Jpg / Png MP4/ MOV format	50%